

Manikandan Kannan

Quality Architect | Playwright Automation Lead | Data Validation Specialist

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PROFESSIONAL SUMMARY

- **11+ years** of technology experience with **7+ years** specializing in enterprise-scale Test Automation Architecture, ETL/Data Validation, and Quality Engineering transformation initiatives.
- Expertise in architecting highly scalable, TypeScript-based **Playwright** frameworks from the ground up, unifying Web, API, and Data testing ecosystems into single-repository solutions.
- Built end-to-end Playwright tests that validate data across the entire system—comparing what the user sees on the screen directly against the backend database records.
- Extensive experience driving enterprise automation using Tricentis **Tosca Commander** and **Tosca Data Integrity (DI)**, including building complex data validation profiles, reusable test assets, and executing large-scale, distributed ETL validation pipelines via **Tosca DEX**.
- Proven track record in driving automation modernization initiatives, improving regression automation coverage from **35% to 80%**, reducing sprint-level manual effort by **40%**, and accelerating execution through distributed validation pipelines.
- Experienced in leading cross-functional QA and Automation teams, driving technical strategy, governance, stakeholder collaboration, and enterprise delivery excellence.
- Skilled in AI-assisted QA engineering and innovation initiatives involving **GitHub Copilot, Claude, MCP**, synthetic test data generation, intelligent evidence management, and enterprise productivity tooling.
- Demonstrated success in managing a 4-member automation team, driving technical mentorship, and ensuring the delivery of scalable, high-quality QA solutions.
- Extensive domain experience across **Financial Audit, Insurance, and US Healthcare** validation projects.

AREAS OF EXPERTISE

Playwright Framework Architecture | TypeScript & Node.js | Shift-Left & In-Sprint Automation |
AI-Driven QA | Web, API & Desktop Automation | CI/CD Pipeline Integration |
ETL & Data Validation | QA Strategy & Leadership | Tosca Commander & DI Automation

CORE TECHNOLOGIES

Primary Stack: Playwright, TypeScript, Node.js, Azure Pipelines, Git, GitHub Copilot, Claude AI, MCP Server

Secondary & Data Stack: WebdriverIO, Appium, Tricentis Tosca, Power BI, Databricks, Snowflake, SQL Server, DAX Studio, Python, SQL

TECHNICAL SKILLS

Category	Skills & Ecosystems
Automation & QA Technologies	Playwright, WebdriverIO, Appium, Tricentis Tosca, Tosca DI, Web Automation, API Automation, Desktop Automation, ETL Validation, Data Validation
Data Engineering & Databases	Databricks, Snowflake, SQL Server, Oracle, Netezza, Informatica PowerCenter, DAX Studio
BI & Reporting Validation	Power BI, Semantic Validation, Metadata Validation, KPI Validation, Source-to-Target Reconciliation
Programming & Scripting	SQL, TypeScript, Python, DAX, Node.js
DevOps & Tools	Azure Pipelines, Git, Jira, DBeaver, PuTTY, Agile/Scrum
AI & Productivity Tools	GitHub Copilot, Claude AI, MCP Server

PROFESSIONAL EXPERIENCE

Quality Architect | NewVision Software *[March 2024 – Present]*

Client/Project: *Deloitte – DataLens (Audit Domain)*

Project Description: An enterprise reporting and data access platform designed for global external audit workflows. The platform enables auditors across multiple countries and member firms to ingest underlying financial data and cross-verify statement accuracy using three core reporting streams: Standard Reporting, Custom Reporting (Power BI Dataflows), and Systemic Data Access.

Key Responsibilities & Achievements:

- Led a 4 member QA Automation team across agile delivery squads, driving test strategy, sprint execution, stakeholder reporting, and technical mentoring.
- Architected a scalable hybrid automation framework using Playwright, Node.js, WebdriverIO, and Appium for unified Web, Desktop, API, and Data Validation.
- Modernized legacy Tosca DI assets into a custom-coded enterprise automation framework, improving scalability, maintainability, and execution efficiency.
- Automated enterprise Power BI semantic validation including DAX KPIs, filters, measures, metadata, dimensions, and reporting layer reconciliation.
- Developed automated DAX Studio validation workflows to reconcile large-scale datasets between backend databases and dashboard outputs.
- Enhanced desktop automation capabilities for complex Windows applications including row-level reconciliation and metadata validation.
- Introduced Shift-Left and In-Sprint automation strategies enabling early validation of previously non-automatable features and reducing regression cycle dependency.

Test Automation Lead | Accenture *[April 2019 – March 2024]*

Client/Project: Nationwide Mutual Insurance Company (*Insurance Domain*)

Project Description: An enterprise data migration initiative involving massive infrastructure modernization moving historical and active policy portfolios from Netezza data store networks over to Snowflake cloud infrastructure using Informatica mappings.

Key Responsibilities & Achievements:

- Validated complex ETL mappings and transformations across Snowflake, Netezza, SQL Server, and Oracle environments, ensuring 99.9% data integrity for migration of 15M+ records.
- Performed large-scale source-to-target reconciliation using advanced SQL validations including row count checks, duplicate detection, aggregation validation, and null verification.
- Built reusable Tosca DI automation assets including test sheets, business attributes, and automated validation profiles, accelerating baseline testing cycles by 45%.
- Managed end-to-end testing delivery including requirement analysis, execution governance, stakeholder communication, and release coordination.
- Implemented distributed ETL validation pipelines using Tosca DEX and Azure Pipelines, reducing execution timelines from days to hours through parallel execution.

Innovation & GenAI Initiatives (POCs)

Airlock Kiosk Automation Solution (Client-Approved POC):

- Delivered a client-approved automation POC for complex Airlock kiosk systems using Windows Registry Autologon and Task Scheduler integration.
- Enabled reliable persistent-session automation for stable non-interactive execution environments.

Low-Code Project Delivery Health Dashboard (Self-Initiated Practice-Wide Solution):

- Designed and implemented a PowerApps-based operational dashboard to track project health, delivery metrics, and resource visibility across the practice.
- Enabled leadership teams with real-time operational insights and centralized reporting for strategic decision-making.

EvidenceIQ – AI-Powered Screenshot Automation & Evidence Management (Self-Initiated POC):

- Developed an AI-driven enterprise QA evidence management POC focused on automated screenshot capture, defect evidence generation, repro-step creation, and trace visualization.
- Integrated Azure DevOps workflows to improve end-to-end validation traceability, defect reporting quality, and functional testing productivity across enterprise applications.

AI-Driven Synthetic Test Data Generator App (Self-Initiated POC - Ongoing):

- Developing an AI-driven synthetic test data generation solution using Playwright, MCP, GitHub Copilot, and Claude AI.
- Enabled schema-based synthetic data generation, intelligent evidence captures, and productivity-focused automation workflows for enterprise QA teams.

TestMind AI (Self-Initiated POC - Ongoing):

- Developed an autonomous AI SDET agent within GitHub Copilot that fully automates agile artifact creation, Playwright execution, and Azure DevOps evidence mapping.
- Powered by a self-updating System Under Test (SUT) knowledge base, it delivers highly accurate, context-aware test automation in a dedicated, cross-platform workspace.

CONSOLIDATED EARLY CAREER EXPERIENCE

Allianz Technology *[June 2016 – April 2019]*

- Performed Manual Testing for insurance backend systems as the primary IT contribution, ensuring strict adherence to technical requirements, stability, and overall accuracy.

Navigant BPM | Pradot | AGS Health *[Jan 2015 – May 2016]*

- Conducted User Acceptance Testing (UAT) for benefit verification systems, provider denial claims software, validating system behavior and ensuring applications met business requirements before production rollout

KEY ACHIEVEMENTS

- Improved regression automation coverage from 35% to 80% through scalable framework modernization initiatives.
- Reduced sprint-level manual testing effort by 40% using Shift-Left and In-Sprint automation strategies.
- Achieved significant execution acceleration through distributed ETL validation pipelines using Tosca DEX and Azure Pipelines.
- Automated enterprise Power BI semantic validation including KPI, metadata, and reconciliation workflows.

CERTIFICATIONS

- Tosca Automation Specialist (AS1 & AS2),
- Tosca Test Design Specialist (TDS1 & TDS2)
- Tosca Automation Engineer (AE1)
- GitHub Copilot Certification (GH-300) (*Certification# Q40603-F8D002*)

EDUCATION

- **B.Tech in Electronics and Communication Engineering (ECE)** - *SASTRA University*